

3rd Exercise Sheet

TUTORIUM: Friday, 8.05.2026.

05

SECTION

Fermi liquid (FL) parameters from a microscopic theory

● PROBLEM

In interacting microscopic theories one of the most important quantities is the self-energy $\Sigma(\mathbf{k}, \omega)$. In the lecture it was shown that for a metal one can connect values extracted out of a Taylor expansion of the self-energy around the Fermi level to the phenomenological parameters of the Landau Fermi liquid theory.

In particular, if the self-energy out of the microscopic theory considered (e.g. dynamical mean-field theory) is purely local [i.e., not $\mathbf{k}$ -dependent: $\Sigma(\mathbf{k}, \omega) = \Sigma(\omega)$], one obtains the following relations:

$$Z = \left[1 - \frac{\partial \operatorname{Re} \Sigma(\omega)}{\partial \omega} \Big|_{\omega \rightarrow 0} \right]^{-1}, \quad \frac{m^*}{m} = Z^{-1}, \quad \Gamma = -\operatorname{Im} \Sigma(\omega)|_{\omega \rightarrow 0} \quad (1)$$

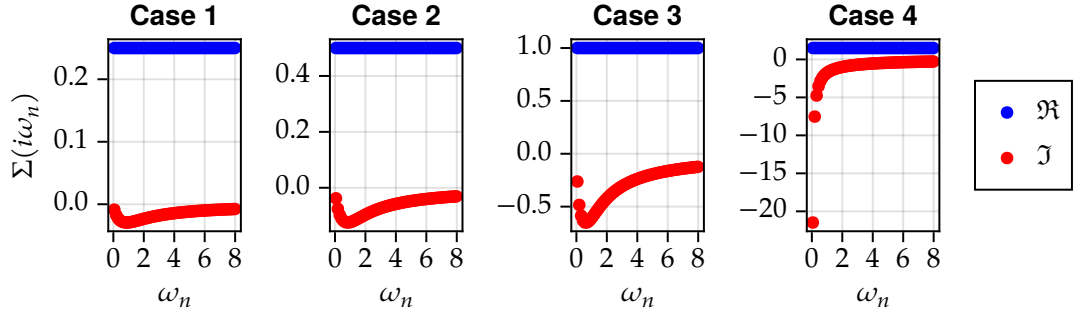
where Z is the quasiparticle weight, m^* the effective mass, m is the bare mass, and Γ defines the quasiparticle scattering rate so that the quasiparticle lifetime is $\tau = (2Z\Gamma)^{-1}$.

Now consider the file `siw.txt` (to be downloaded from TISS). It contains local fermionic self-energies for positive Matsubara frequencies $\Sigma_l(i\omega_n)$ for four different physical cases $l = 1, 2, 3, 4$ at the inverse temperature $\frac{1}{T} \equiv \beta = 50 \text{ eV}^{-1}$. The columns of the file contain the following: $\omega_n \mid \operatorname{Re} \Sigma_1(i\omega_n) \mid \operatorname{Im} \Sigma_1(i\omega_n) \mid \operatorname{Re} \Sigma_2(i\omega_n) \cdots$

a)

● PROBLEM
SOLUTION

Plot the real and imaginary parts of the self-energies in all cases, respectively.



b)

PROBLEM

For cases one to three numerically extract the quasiparticle weight Z , the effective mass $\frac{m^*}{m}$ and Γ . What is the difference between these cases and how could one interpret those? Hint: In principle one would have to analytically continue the self-energy from the Matsubara to the real frequency axis ($i\omega_n \rightarrow \omega + i\delta$). Show that these parameters can also be approximately obtained directly from values at Matsubara frequencies as follows:

$$Z = \left[1 - \frac{\partial \text{Im} \Sigma(i\omega_n)}{\partial \omega_n} \Big|_{\omega_n \rightarrow 0^+} \right]^{-1}, \quad \Gamma = -\text{Im} \Sigma(i\omega_n) \Big|_{\omega_n \rightarrow 0^+} \quad (2)$$

SOLUTION

Assuming that $\Sigma(z)$ is analytic near $z = 0$, we expand

$$\Sigma(z) = \Sigma(0) + z\Sigma'(0) + \mathcal{O}(|z|^2). \quad (3)$$

On the real-frequency axis, $z = \omega$, hence

$$\Sigma(\omega) = \Sigma(0) + \omega\Sigma'(0) + \mathcal{O}(\omega^2), \quad (4)$$

so

$$\text{Re} \Sigma(\omega) = \text{Re} \Sigma(0) + \omega \text{Re} \Sigma'(0) + \mathcal{O}(\omega^2), \quad (5)$$

and therefore

$$\frac{\partial \text{Re} \Sigma(\omega)}{\partial \omega} \Big|_{\omega \rightarrow 0} = \text{Re} \Sigma'(0). \quad (6)$$

On the Matsubara axis, $z = i\omega_n$, hence

$$\Sigma(i\omega_n) = \Sigma(0) + i\omega_n \Sigma'(0) + \mathcal{O}(\omega_n^2), \quad (7)$$

which implies

$$\text{Im} \Sigma(i\omega_n) = \text{Im} \Sigma(0) + \omega_n \text{Re} \Sigma'(0) + \mathcal{O}(\omega_n^2). \quad (8)$$

Thus,

$$\left. \frac{\partial \text{Im } \Sigma(i\omega_n)}{\partial \omega_n} \right|_{\omega_n \rightarrow 0^+} = \text{Re } \Sigma'(0) = \left. \frac{\partial \text{Re } \Sigma(\omega)}{\partial \omega} \right|_{\omega \rightarrow 0}. \quad (9)$$

Inserting this into Eq. Equation 1 gives

$$Z = \left[1 - \left. \frac{\partial \text{Im } \Sigma(i\omega_n)}{\partial \omega_n} \right|_{\omega_n \rightarrow 0^+} \right]^{-1}. \quad (10)$$

Moreover, taking $\omega_n \rightarrow 0^+$ yields

$$\text{Im } \Sigma(i\omega_n)|_{\omega_n \rightarrow 0^+} = \text{Im } \Sigma(0), \quad (11)$$

so, using

$$\Gamma = -\text{Im } \Sigma(\omega)|_{\omega \rightarrow 0}, \quad (12)$$

one obtains the Matsubara estimate

$$\Gamma \approx -\text{Im } \Sigma(i\omega_n)|_{\omega_n \rightarrow 0^+}. \quad (13)$$

Hence, if the first Matsubara frequencies already lie in the low-frequency regime, both Z and Γ can be extracted approximately directly from Matsubara data.

Extraction from the data

To compute the Fermi-liquid parameters, we need the low-frequency behavior of the Matsubara self-energy. In particular,

$$Z = \frac{1}{1 - \left. \frac{\partial \text{Im } \Sigma(i\omega_n)}{\partial \omega_n} \right|_{\omega_n \rightarrow 0^+}}, \quad \Gamma = -\text{Im } \Sigma(i\omega_n)|_{\omega_n \rightarrow 0^+}. \quad (14)$$

Since the data is only available at discrete Matsubara frequencies, the zero-frequency limit has to be obtained by extrapolation. For each case, we fit the first N Matsubara points of the imaginary part of the self-energy by a polynomial

$$\text{Im } \Sigma(i\omega_n) \approx a_0 + a_1\omega_n + a_2\omega_n^2 + \dots \quad (15)$$

The fitted intercept and slope then give

$$\Gamma = -a_0, \quad Z = \frac{1}{1 - a_1}, \quad \frac{m^*}{m} = \frac{1}{Z}. \quad (16)$$

To estimate the systematic uncertainty of this extrapolation, we repeat the fit for different polynomial orders and for different numbers of included low-frequency Matsubara points. The matrices below show the resulting stability scan.

For the final quoted values, we use the median over the relatively stable low-frequency block with polynomial orders 2 and 3 and $N = 4, 5, 6$ included Matsubara points. The quoted uncertainty is half the range over this selected block,

$$\Delta X = \frac{\max(X) - \min(X)}{2}, \quad (17)$$

where X denotes Z , $\frac{m^*}{m}$, or Γ . This uncertainty should therefore be interpreted as a systematic uncertainty from the extrapolation, not as a statistical error.

Case	Z	$\frac{m^*}{m}$	Γ
1	0.91 ± 0.01	1.1 ± 0.02	0.0032 ± 0.001
2	0.71 ± 0.04	1.4 ± 0.08	0.014 ± 0.005
3	0.28 ± 0.06	3.5 ± 0.7	0.12 ± 0.04

The three cases show increasing correlation strength. From case 1 to case 3, the quasiparticle weight Z decreases, while the effective mass $\frac{m^*}{m} = \frac{1}{Z}$ increases. This means that the quasiparticles become heavier and less coherent. At the same time, the extrapolated scattering rate Γ increases, indicating a shorter quasiparticle lifetime. Thus case 1 is closest to a weakly correlated Fermi liquid, case 2 is a more strongly renormalized Fermi liquid, and case 3 is close to the breakdown of the Fermi-liquid picture.

Case 1: Z estimates

order	3	4	5	6	7	8	9	10
1	0.95	0.96	0.96	0.97	0.98	0.98	0.98	0.99
2	0.91	0.92	0.92	0.93	0.94	0.94	0.95	0.95
3	–	0.9	0.91	0.91	0.91	0.92	0.92	0.92

Case 1: $\frac{m^*}{m}$ estimates

order	3	4	5	6	7	8	9	10
1	1.1	1	1	1	1	1	1	1
2	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
3	–	1.1	1.1	1.1	1.1	1.1	1.1	1.1

Case 1: Γ estimates

order	3	4	5	6	7	8	9	10
1	0.0058	0.0073	0.0088	0.01	0.012	0.013	0.014	0.016
2	0.003	0.0035	0.004	0.0047	0.0053	0.006	0.0067	0.0074
3	–	0.0025	0.0027	0.003	0.0032	0.0035	0.0038	0.0042

Case 2: Z estimates

order	3	4	5	6	7	8	9	10
1	0.81	0.84	0.86	0.89	0.9	0.92	0.93	0.94
2	0.7	0.72	0.74	0.76	0.77	0.79	0.8	0.81
3	–	0.68	0.69	0.7	0.71	0.72	0.73	0.74

Case 2: $\frac{m^*}{m}$ estimates

order	3	4	5	6	7	8	9	10
1	1.2	1.2	1.2	1.1	1.1	1.1	1.1	1.1
2	1.4	1.4	1.3	1.3	1.3	1.3	1.2	1.2
3	–	1.5	1.5	1.4	1.4	1.4	1.4	1.4

Case 2: Γ estimates

order	3	4	5	6	7	8	9	10
1	0.025	0.032	0.039	0.045	0.051	0.057	0.063	0.068
2	0.013	0.015	0.018	0.021	0.023	0.026	0.029	0.032
3	–	0.011	0.012	0.013	0.015	0.016	0.017	0.019

Case 3: Z estimates

order	3	4	5	6	7	8	9	10
1	0.44	0.51	0.58	0.64	0.7	0.75	0.8	0.85
2	0.27	0.3	0.33	0.36	0.39	0.42	0.45	0.48
3	–	0.24	0.25	0.27	0.28	0.3	0.31	0.33

Case 3: $\frac{m^*}{m}$ estimates

order	3	4	5	6	7	8	9	10
1	2.3	2	1.7	1.6	1.4	1.3	1.2	1.2
2	3.7	3.4	3.1	2.8	2.6	2.4	2.2	2.1
3	–	4.2	4	3.7	3.5	3.4	3.2	3

Case 3: Γ estimates

order	3	4	5	6	7	8	9	10
1	0.2	0.25	0.29	0.33	0.37	0.4	0.43	0.46
2	0.11	0.13	0.15	0.17	0.2	0.22	0.24	0.26
3	–	0.087	0.098	0.11	0.12	0.14	0.15	0.16

c)

PROBLEM • SOLUTION

What about the fourth case? See also exercise 6.

Case 4 is qualitatively different from the first three cases. In the plot, the real part is essentially constant,

$$\text{Re } \Sigma_4(i\omega_n) \approx 1.5, \quad (18)$$

while the imaginary part becomes very large and negative as $\omega_n \rightarrow 0^+$. This behavior is not compatible with a regular Fermi-liquid Taylor expansion around zero frequency.

Instead, the data suggests a pole-like low-frequency self-energy,

$$\Sigma_4(i\omega_n) \approx \Sigma_0 + \frac{C}{i\omega_n} = \Sigma_0 - i\frac{C}{\omega_n}, \quad (19)$$

with $C > 0$. Therefore,

$$\text{Im } \Sigma_4(i\omega_n) \approx -\frac{C}{\omega_n}. \quad (20)$$

This directly explains the strong downturn of the red data points in the plot. It also implies

$$-\text{Im } \Sigma_4(i\omega_n) \approx \frac{C}{\omega_n} \rightarrow \infty \quad \text{for } \omega_n \rightarrow 0^+. \quad (21)$$

Thus the Fermi-liquid estimate of the scattering rate does not give a finite value. More importantly, the derivative entering the quasiparticle weight diverges,

$$\frac{\partial \text{Im } \Sigma_4(i\omega_n)}{\partial \omega_n} \approx \frac{C}{\omega_n^2} \rightarrow \infty, \quad (22)$$

and hence

$$Z = \frac{1}{1 - \left. \frac{\partial \text{Im } \Sigma(i\omega_n)}{\partial \omega_n} \right|_{\omega_n \rightarrow 0^+}} \rightarrow 0. \quad (23)$$

Consequently,

$$\frac{m^*}{m} = Z^{-1} \rightarrow \infty. \quad (24)$$

Therefore case 4 does not describe a Fermi liquid. The quasiparticle weight vanishes, the effective mass diverges, and there is no well-defined finite quasiparticle scattering rate.

This is consistent with the atomic-limit Hubbard model discussed in exercise 6. At half filling one obtains

$$\Sigma(i\omega_n) = \frac{U}{2} + \frac{U^2}{4i\omega_n} = \frac{U}{2} - i\frac{U^2}{4\omega_n}. \quad (25)$$

This has exactly the same structure as case 4: a constant real part and a $\frac{1}{\omega_n}$ divergence in the imaginary part. The constant real part is consistent with the atomic-limit form up to the precise value of the interaction parameter and possible convention-dependent shifts. The important point is the qualitative structure: a constant real part and a $\frac{1}{\omega_n}$ divergence in the imaginary part.

Thus case 4 can be interpreted as a strongly correlated Mott-insulating or atomic-limit-like case. The pole in the self-energy opens a gap at the Fermi level and destroys Landau quasiparticles.

In summary,

$$Z = 0, \quad \frac{m^*}{m} \rightarrow \infty, \quad \Gamma \text{ is not finite / not a well-defined FL scattering rate.} \quad (26)$$

A simple model with strong correlations

PROBLEM

Landau FL theory, as a theory for metals, reproduces many qualitative features of the non-interacting Fermi gases. In such systems, the screened Coulomb energy is much smaller than the electronic kinetic energy. As a result, the electron-electron interactions are effectively small perturbations to the motion of electrons at the Fermi level.

In another limit, i.e. when Coulomb energy is much stronger than the electronic kinetic energy, FL theory does not hold anymore. An intuitive, yet simple, example for understanding the complete disappearance of quasiparticles due to the strong electron-electron correlations is the Hubbard model at the atomic limit, where it reads:

$$H_{\text{at}} = U n_{\uparrow} n_{\downarrow} - \mu (n_{\uparrow} + n_{\downarrow}) \quad (27)$$

Let us consider here the half-filling case, i.e. one electron per site, obtained for chemical potential $\mu = \frac{U}{2}$ which we absorb in the Hamiltonian. We observe that there is only one energy scale in this model, the local Coulomb repulsion U . As the particle numbers $n_{\uparrow}$ and $n_{\downarrow}$ are good quantum numbers of this Hamiltonian, the basis of the problem can be simply taken as the particle number basis:

$$|0\rangle, \quad |\uparrow\rangle, \quad |\downarrow\rangle, \quad |\uparrow\downarrow\rangle, \quad (28)$$

which represents the empty, singly occupied (up/down) and doubly occupied states, correspondingly.

a)

PROBLEM

Show that the states in Equation 28 are also the eigenstates of H_{at} and find the corresponding eigenenergies. Show that the partition function at inverse temperature β is given as $\mathcal{Z} = \text{tr } e^{-\beta H} = 2(1 + e^{\beta U/2})$.

SOLUTION

There holds

$$\begin{aligned} n_{\sigma} |0\rangle &= 0 \\ n_{\sigma} |\sigma\rangle &= |\sigma\rangle \\ n_{\sigma} |-\sigma\rangle &= 0 \\ n_{\sigma} |\uparrow\downarrow\rangle &= |\uparrow\downarrow\rangle \end{aligned} \quad (29)$$

and so

$$\begin{aligned}
H_{\text{at}}|0\rangle &= [Un_{\uparrow}n_{\downarrow} - \mu(n_{\uparrow} + n_{\downarrow})]|0\rangle = 0 \\
H_{\text{at}}|\uparrow\rangle &= [Un_{\uparrow}n_{\downarrow} - \mu(n_{\uparrow} + n_{\downarrow})]|\uparrow\rangle = -\mu|\uparrow\rangle \\
H_{\text{at}}|\downarrow\rangle &= [Un_{\uparrow}n_{\downarrow} - \mu(n_{\uparrow} + n_{\downarrow})]|\downarrow\rangle = -\mu|\downarrow\rangle \\
H_{\text{at}}|\uparrow\downarrow\rangle &= [Un_{\uparrow}n_{\downarrow} - \mu(n_{\uparrow} + n_{\downarrow})]|\uparrow\downarrow\rangle = (U - 2\mu)|\uparrow\downarrow\rangle,
\end{aligned} \tag{30}$$

showing that the given states are indeed eigenstates of H_{at} .

The partition function is

$$\begin{aligned}
\mathcal{Z} &= \text{tr } e^{-\beta H} \\
&= \sum_n \langle n | e^{-\beta H} | n \rangle \\
&= \sum_n e^{-\beta E_n} \\
&= e^0 + e^{\beta\mu} + e^{\beta\mu} + e^{-\beta(U-2\mu)}
\end{aligned} \tag{31}$$

and at half-filling $\mu = \frac{U}{2}$

$$\mathcal{Z} = 2(1 + e^{\beta U/2}). \tag{32}$$

b)

PROBLEM

Try to express the creation and annihilation operators $\hat{c}_{\sigma}^{\dagger}$ and $\hat{c}_{\sigma}$ as 4×4 matrices in the above basis. Show that the imaginary-time Green's function is given by:

$$G_{\sigma}(\tau) = -\langle T_{\tau} \hat{c}_{\sigma}(\tau) \hat{c}_{\sigma}^{\dagger} \rangle = -\frac{1}{\mathcal{Z}} \text{tr} [e^{-\beta H} \hat{c}_{\sigma}(\tau) \hat{c}_{\sigma}^{\dagger}] = -\frac{e^{\frac{U}{2}\tau} + e^{(\beta-\tau)\frac{U}{2}}}{\mathcal{Z}}, \tag{33}$$

where σ stands for the spin orientation $\uparrow$ (or $\downarrow$).

SOLUTION

We take

$$|\uparrow\downarrow\rangle \equiv \hat{c}_{\uparrow}^{\dagger} \hat{c}_{\downarrow}^{\dagger} |0\rangle \tag{34}$$

and then get

$$\begin{aligned}
\hat{c}_{\sigma}^{\dagger} |0\rangle &= |\sigma\rangle & \hat{c}_{\sigma} |0\rangle &= 0 \\
\hat{c}_{\sigma}^{\dagger} |\sigma\rangle &= 0 & \hat{c}_{\sigma} |\sigma\rangle &= |0\rangle \\
\hat{c}_{\sigma}^{\dagger} |\uparrow\downarrow\rangle &= 0 & \hat{c}_{\sigma} |-\sigma\rangle &= 0 \\
\hat{c}_{\uparrow}^{\dagger} |\downarrow\rangle &= |\uparrow\downarrow\rangle & \hat{c}_{\uparrow} |\uparrow\downarrow\rangle &= |\downarrow\rangle \\
\hat{c}_{\downarrow}^{\dagger} |\uparrow\rangle &= -|\uparrow\downarrow\rangle & \hat{c}_{\downarrow} |\uparrow\downarrow\rangle &= -|\uparrow\rangle
\end{aligned} \tag{35}$$

In the ordered basis

$$(|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle) \tag{36}$$

we get the representation (with the matrix O representing the operator $\hat{O}$ given by $O_{mn} = \langle m | \hat{O} | n \rangle$)

$$\begin{aligned}
\hat{c}_\uparrow^\dagger &\leftrightarrow \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot \end{pmatrix} & \hat{c}_\downarrow^\dagger &\leftrightarrow \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot \\ \cdot & -1 & \cdot & \cdot \end{pmatrix} \\
\hat{c}_\uparrow &\leftrightarrow \begin{pmatrix} \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} & \hat{c}_\downarrow &\leftrightarrow \begin{pmatrix} \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & -1 \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}
\end{aligned} \tag{37}$$

Bonus:

$$n_\uparrow = \hat{c}_\uparrow^\dagger \hat{c}_\uparrow \leftrightarrow \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 \end{pmatrix} \quad n_\downarrow = \hat{c}_\downarrow^\dagger \hat{c}_\downarrow \leftrightarrow \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & 1 \end{pmatrix} \tag{38}$$

$$H = U n_\uparrow n_\downarrow - \mu (n_\uparrow + n_\downarrow) \leftrightarrow \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & -\mu & \cdot & \cdot \\ \cdot & \cdot & -\mu & \cdot \\ \cdot & \cdot & \cdot & U - 2\mu \end{pmatrix} \tag{39}$$

$$e^{\tau H} \leftrightarrow \begin{pmatrix} 1 & \cdot & \cdot & \cdot \\ \cdot & e^{-\mu\tau} & \cdot & \cdot \\ \cdot & \cdot & e^{-\mu\tau} & \cdot \\ \cdot & \cdot & \cdot & e^{(U-2\mu)\tau} \end{pmatrix} \tag{40}$$

$$\hat{c}_\uparrow(\tau) = e^{\tau H} \hat{c}_\uparrow e^{-\tau H} \leftrightarrow \begin{pmatrix} \cdot & e^{\mu\tau} & \cdot & \cdot \\ \cdot & \cdot & \cdot & e^{(-U+\mu)\tau} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} \tag{41}$$

$$\hat{c}_\downarrow(\tau) = e^{\tau H} \hat{c}_\downarrow e^{-\tau H} \leftrightarrow \begin{pmatrix} \cdot & \cdot & e^{\mu\tau} & \cdot \\ \cdot & \cdot & \cdot & -e^{(-U+\mu)\tau} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

$$e^{-\beta H} \hat{c}_\uparrow(\tau) \hat{c}_\uparrow^\dagger \leftrightarrow \begin{pmatrix} e^{\mu\tau} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & e^{-U\tau+\mu(\beta+\tau)} & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} \tag{42}$$

$$e^{-\beta H} \hat{c}_\downarrow(\tau) \hat{c}_\downarrow^\dagger \leftrightarrow \begin{pmatrix} e^{\mu\tau} & \cdot & \cdot & \cdot \\ \cdot & e^{-U\tau+\mu(\beta+\tau)} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

The Green's function then is

$$\begin{aligned}
G_\sigma(\tau) &= -\langle T_\tau \hat{c}_\sigma(\tau) \hat{c}_\sigma^\dagger \rangle \\
&= -\frac{1}{\mathcal{Z}} \text{tr} [e^{-\beta H} \hat{c}_\sigma(\tau) \hat{c}_\sigma^\dagger] \\
&= -\frac{1}{\mathcal{Z}} (e^{\mu\tau} + e^{-U\tau + \mu(\beta + \tau)}) \\
&= -\frac{e^{\mu\tau} + e^{-U\tau + \mu(\beta + \tau)}}{1 + 2e^{\beta\mu} + e^{\beta(-U + 2\mu)}}
\end{aligned} \tag{43}$$

and in half-filling $\mu = \frac{U}{2}$ this simplifies to

$$G_\sigma(\tau) = -\frac{e^{\frac{U}{2}\tau} + e^{(\beta - \tau)\frac{U}{2}}}{\mathcal{Z}} = -\frac{e^{\frac{U}{2}\tau} + e^{(\beta - \tau)\frac{U}{2}}}{2(1 + e^{\beta\frac{U}{2}})}. \tag{44}$$

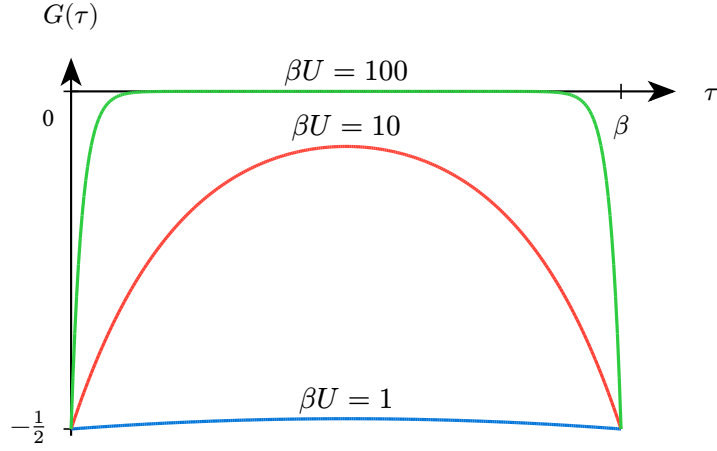


Figure 1: Green's function $G(\tau)$ plotted against τ for several values of βU .

c)

PROBLEM

Fourier transform the imaginary-time Green's function $G_\sigma(\tau)$ to Matsubara space to get $G_\sigma(i\omega_n) = \int_0^\beta G_\sigma(\tau) e^{i\omega_n \tau} d\tau$ and determine the corresponding spectral function $A_\sigma(\omega)$ and the self-energy $\Sigma_\sigma(\omega)$. [Hint: to get $A_\sigma(\omega)$ at real frequency axis, an analytical continuation $i\omega_n \rightarrow \omega + i\delta$ is required.] Does the system have an energy gap, and if so how large is it? Do we have well-defined quasiparticles in this case? [Hint: try to determine the quasiparticle weight Z , effective mass m^* and scattering rate Γ as defined in Equation 1.]

SOLUTION

With

$$n \equiv \langle n_\sigma \rangle = \frac{e^{\beta\mu} + e^{\beta(-U + 2\mu)}}{1 + 2e^{\beta\mu} + e^{\beta(-U + 2\mu)}} \tag{45}$$

the Matsubara Green's function reads

$$\begin{aligned}
G_\sigma(i\omega_n) &= \int_0^\beta G_\sigma(\tau) e^{i\omega_n \tau} d\tau \\
&= \frac{n}{i\omega_n + \mu - U} + \frac{1-n}{i\omega_n + \mu} \\
&= \frac{1}{i\omega_n + \mu - \Sigma_\sigma(i\omega_n)}
\end{aligned} \tag{46}$$

with the self-energy

$$\Sigma_\sigma(i\omega_n) = nU + \frac{n(1-n)U^2}{i\omega_n + \mu - (1-n)U}. \tag{47}$$

At half-filling $\mu = \frac{U}{2}$, $n = \frac{1}{2}$, and the Green's function simplifies to

$$G_\sigma(i\omega_n) = \frac{\frac{1}{2}}{i\omega_n - \frac{U}{2}} + \frac{\frac{1}{2}}{i\omega_n + \frac{U}{2}} = \frac{1}{i\omega_n - \frac{U^2}{4i\omega_n}} \tag{48}$$

with self-energy

$$\Sigma_\sigma(i\omega_n) = \frac{U}{2} + \frac{U^2}{4i\omega_n}. \tag{49}$$

The spectral function is

$$\begin{aligned}
A_\sigma(\omega) &= -\frac{1}{\pi} \text{Im} G_\sigma(\omega + i\delta) \\
&= nL_{U-\mu}(\omega) + (1-n)L_{-\mu}(\omega),
\end{aligned} \tag{50}$$

with the Lorentzian

$$L_{x_0}(x) = \frac{1}{\pi} \frac{\delta}{(x - x_0)^2 + \delta^2}. \tag{51}$$

For plots, we use $\Delta\mu = \mu - \frac{U}{2}$ and set $\beta U = 1$ and $\beta\delta = 10^{-9}$.

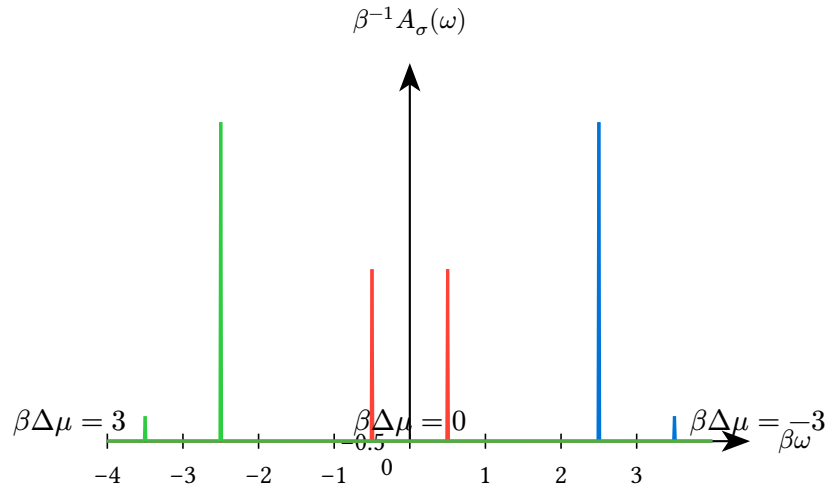


Figure 2: Spectral function $A_\sigma(\omega)$ plotted against ω for several values of $\beta\Delta\mu$.

The self-energy in real frequency is

$$\Sigma_{\sigma}(\omega) = nU + \frac{n(1-n)U^2}{\omega + i\delta + \mu - (1-n)U} \quad (52)$$

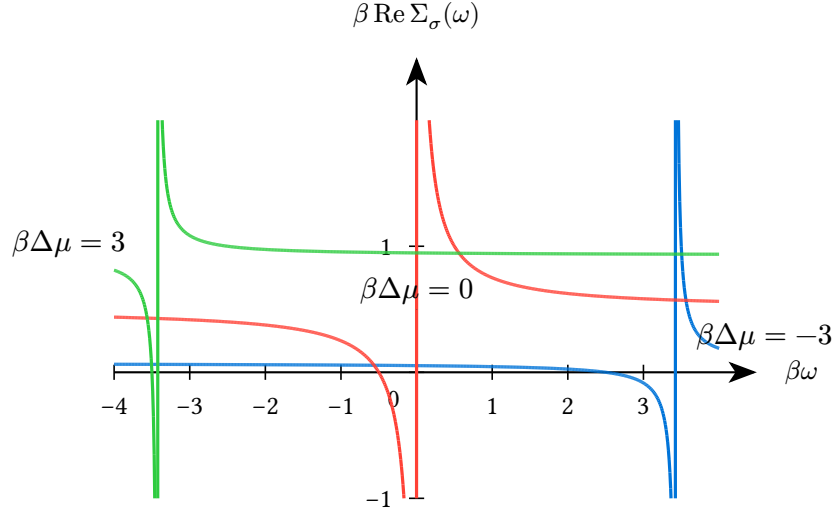


Figure 3: Real part of the self-energy $\text{Re } \Sigma_{\sigma}(\omega)$ plotted against ω for several values of $\beta\Delta\mu$.

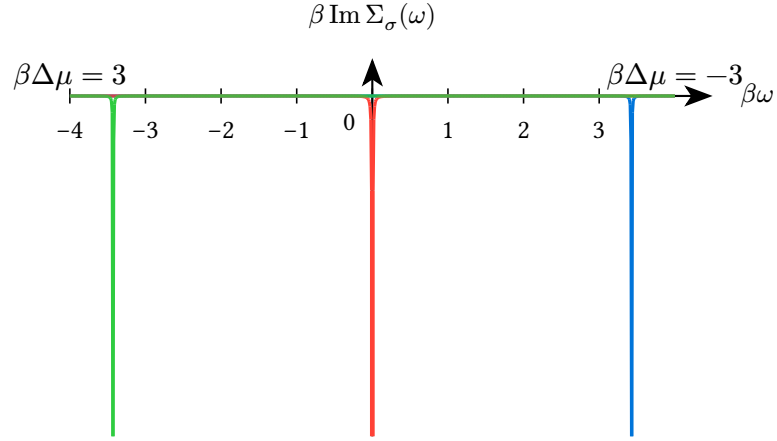


Figure 4: Imaginary part of the self-energy $\text{Im } \Sigma_{\sigma}(\omega)$ plotted against ω for several values of $\beta\Delta\mu$.

In the limit $\delta \rightarrow 0^+$, the Lorentzians become delta distributions. For $0 < \mu < U$, the Fermi level lies in the gap between the two Hubbard peaks. The full spectral gap between the two delta peaks is

$$\Delta_{\text{gap}} = U, \quad (53)$$

but the distance from the Fermi level to the nearest spectral excitation is

$$\Delta_{\text{exc}} = \min(\mu, U - \mu). \quad (54)$$

Without broadening, $\text{Re } \Sigma(\omega) = nU + \frac{n(1-n)U^2}{\omega + \mu - (1-n)U}$ and we get quasiparticle weight

$$\begin{aligned} Z &= \left[1 - \frac{\partial \text{Re } \Sigma(\omega)}{\partial \omega} \Big|_{\omega \rightarrow 0} \right]^{-1} \\ &= \frac{(\mu - (1-n)U)^2}{\mu^2 - 2\mu(1-n)U + (1-n)U^2}. \end{aligned} \quad (55)$$

Also without broadening, $\text{Im } \Sigma(\omega) = -\pi(1-n)nU^2\delta(\omega + \mu - (1-n)U)$, so the quasiparticle scattering rate is

$$\begin{aligned}\Gamma &= -\text{Im } \Sigma(\omega)|_{\omega \rightarrow 0} \\ &= \pi(1-n)nU^2\delta(\mu - (1-n)U).\end{aligned}\tag{56}$$

At half-filling, $\mu = \frac{U}{2}$ and $n = \frac{1}{2}$, so

$$Z = 0 \quad \text{and} \quad \frac{m^*}{m} = Z^{-1} \rightarrow \infty \quad \text{and} \quad \Gamma \rightarrow \infty\tag{57}$$

and the system is a gapped atomic-limit Mott insulator with no well-defined quasiparticles.